AI & Machine Learning for Business

Week 8 • Support Vector Machines, Model Selection & BookingGuard

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TODAY'S DECISION

When several models fit the past equally well, which one should we trust with a policy decision?

Course status and required notices

COURSE PROGRESS	Weeks 1–7 built probability, validation, regularization, nonlinear features, trees, forests and boosting. Today adds the last classifier family before the midterm, then turns it into a model-selection habit.
SAVE OR LET GO	Closed Wednesday, October 14. Confirm your Blackboard receipt today. Its threshold, sensitivity and memo habits return in BookingGuard.
MIDTERM	Friday, October 30, in the 3:00–6:00 PM class block. Sessions 1–8, so today's SVM content is in scope.
CAPSTONE	Interest Exchange is complete. Register a proposed team by Tuesday, October 27, 11:59 PM ET. Teams become official only after instructor confirmation on October 28.
BOOKINGGUARD	Exercise 03 launches today. Due Sunday, November 15, 11:59 PM ET. Individual exercise. It applies today's model-selection framework to a deposit policy.

Blackboard is the home for everything this week

When a slide shows the teal Blackboard tag in its lower right corner, that is where to go.

WEEK 8 PAGE	Slides (PDF), the in-class SVM workbook (opens in Colab), and the retrieval check
EXERCISE 02	Save or Let Go: closed October 14. Your submission receipt is under Exercises › Save or Let Go
EXERCISE 03	BookingGuard: brief, starter notebook, policy_costs.csv, rubric and submission field. Due November 15
CAPSTONE AREA	01. Assessment & Grading › Group Capstone Project: team registration, confirmed teams, proposal template
MIDTERM	Friday, October 30. Format and access instructions are posted on Blackboard

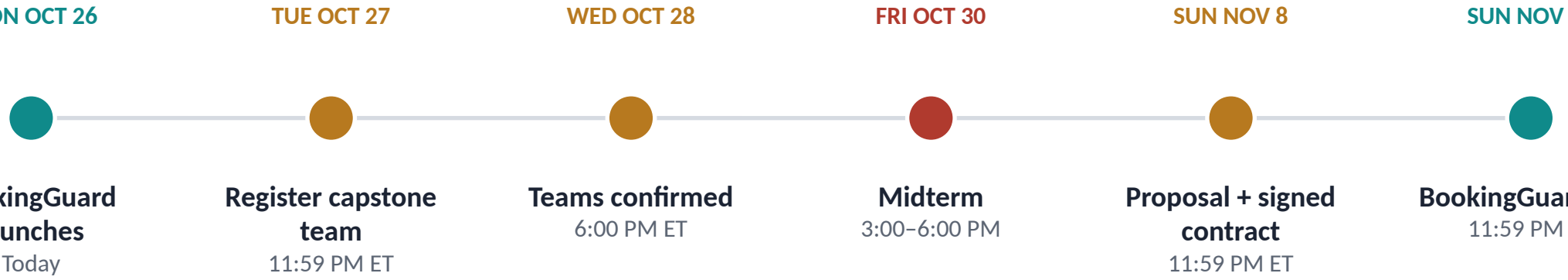
Workbooks: open from Blackboard, then File › Save a copy in Drive. Work in your copy; the original stays read-only.

Three hours: build the foundation, add the SVM, then put it to work

TIME	BLOCK	BY THE END YOU CAN...
0–10 min	Opening, notices, Save or Let Go closeout	Name this week's deadlines and confirm your Exercise 02 receipt
10–25 min	Midterm: format and one example of each question type	Answer each question type the way it is scored
25–45 min	Foundations: bias–variance and the Week 7 bridge	Place the SVM among the classifiers you already know
45–90 min	SVM: why → margin → soft margin → kernels → probabilities	Explain margin, C, scaling, gamma and calibration
90–105 min	*Break*	
105–130 min	In-class SVM workbook, Tasks 1–5	Tune and judge an SVM with validation evidence
130–150 min	Model-selection framework	Defend a model choice as a business decision
150–168 min	BookingGuard launch	State the prediction moment and a first hypothesis
168–180 min	Capstone proposals and exit ticket	Recognize a strong proposal before November 8

Six dates between today and November 15

Three arrive before Friday. Put them in your calendar now.



BEFORE FRIDAY

Register your team, confirm the Save or Let Go receipt, and finish your one double-sided midterm reference sheet.

AFTER FRIDAY

Proposal and contract by November 8; BookingGuard by November 15. Start BookingGuard's first hypothesis tonight.

Confirm the submission, then carry its habits into BookingGuard

THE ASK · TODAY

1 Open Blackboard › Exercises › Save or Let Go.

2 Confirm the receipt lists your memo and methodology deck.

3 If a file or receipt is missing, raise it with the instructor today, not after the midterm.

4 Keep your threshold and sensitivity code: BookingGuard reuses the pattern.

WHAT CARRIES FORWARD

IN SAVE OR LET GO	RETURNS IN BOOKINGGUARD AS
Threshold from campaign economics, not 0.50	Deposit threshold from policy_costs.csv
Validation that represents future use	Train on earlier arrivals, validate on a later period
Sensitivity to contact cost and retained value	Sensitivity to deposit recovery and guest friction
Contact volume under a capacity limit	Deposits per 1,000 bookings
One-page memo: who acts, how many, value, limitation	One-page memo plus a decision file

BLACKBOARD › Exercises › Save or Let Go

Midterm first: the final preparation window

105	15	3	1
minutes	multiple-choice concept checks	applied written responses	integrated written case

RULES	One double-sided reference sheet and a basic calculator. No AI tools, internet, collaboration or coding.
SCOPE	Sessions 1–8. BookingGuard and capstone logistics are excluded.
CONCEPT CHECKS	Multiple choice: one best answer from four. Distractors are common misconceptions.
WRITTEN ITEMS	Calculation or evidence with units → decision → the assumption that would reverse it.

Concept check: choose the one best answer

Commit to a letter in 60 seconds, then check the key. A practice item, not an exam item.

Fraud is 1% of transactions. A new model scores 99% accuracy on validation data. What should you conclude?

A The model is excellent

B The model is overfitting

C Accuracy alone is uninformative here; compare precision, recall or PR-AUC with the 1% baseline

D The data must contain leakage

Answer C. Predicting “no fraud” for everyone also scores 99%, so accuracy cannot separate a useful model from a useless one.

Classification threshold: show the calculation, then the reversal

Written prompt · A retention model assigns p(churn) = 0.40. An intervention costs $20 and saves $80 when churn would occur. Should the team intervene? State one assumption that could reverse your answer.

MODEL ANSWER

CALCULATE

Expected recovery = 0.40 × $80 = $32

Net expected value = $32 − $20 = +$12

Break-even p* = $20 ÷ $80 = 0.25

DECIDE

Intervene: 0.40 is above the 0.25 break-even, under these assumptions.

Units: dollars per customer contacted.

REVERSAL

If the offer works only half the time: 0.40 × 0.5 × $80 = $16 < $20.

Then do not intervene. Break-even success rate: 62.5%.

Show units, compare value with cost, connect the arithmetic to the action, then name the assumption that flips it.

Audit an end-to-end modeling workflow

Written prompt · A subscription team predicts churn 30 days before renewal with gradient boosting. They target-encode the plan field using all 2023–2025 rows, split randomly 80/20, check test AUC after every tuning run, include a field recording whether a retention offer was accepted, and act when p ≥ 0.50.

EVIDENCE BREAK	WHY IT MISLEADS	REPAIR
Offer-accepted field	Known only after the decision: target leakage	Keep only fields known 30 days before renewal
Encoding on all rows; random split	Validation labels shape features; mixes years	Fit encoding on earlier rows; validate on a later period
Test AUC after every run	The test set becomes tuning data: optimistic	Tune on validation; touch the test set once
Acting at p ≥ 0.50	A habit, not a policy	Break-even from offer cost and saved value; stress-test it

Build your one double-sided reference sheet before Friday

FORMULAS

- Expected value and break-even p* = cost ÷ value saved
- Precision, recall, F1
- Log loss and Brier score
- Expected cost under a capacity limit

DIAGNOSTICS

- Train vs validation patterns for bias and variance
- Leakage test: known at the prediction moment?
- Capacity knob for each model family
- Validation-curve stopping rule

DECISIONS

- Random vs temporal validation
- Model comparison: same rows, split, metrics
- Threshold under capacity
- Recommendation + reversal trigger

WEEK 8 · PART 1

The classifier toolkit so far

Seven building blocks, one lens, and the question Week 7 left open

Seven building blocks you already own

BUILDING BLOCK	WHAT IT GAVE US	THE QUESTION IT ANSWERS
Probability & thresholds	Scores become actions through costs	When is a prediction worth acting on?
Validation	Evidence that represents future use	Will this work on next month's data?
Regularization	A penalty that trades fit for stability	How do we keep a flexible model stable?
Nonlinear features	Curved boundaries by adding features	What if a straight boundary is not enough?
Trees	Piecewise rules; feature scale does not matter	Can we capture interactions without formulas?
Forests	Averaging cuts variance	How do we steady an unstable model?
Boosting	Loss + gradient steps + capacity control	How do we fix what the model still gets wrong?

Today's new model reuses every one of these blocks. We will name each link once the model is on the table.

The target and the decision identify the learning problem

Today stays inside classification: an event or category that triggers an action.

REGRESSION

Predict a number

Revenue, demand, length of stay

CLASSIFICATION

Predict an event or category

Churn, fraud, cancellation: today's focus

CLUSTERING

Discover groups with no label

Customer segments for a campaign

REDUCTION

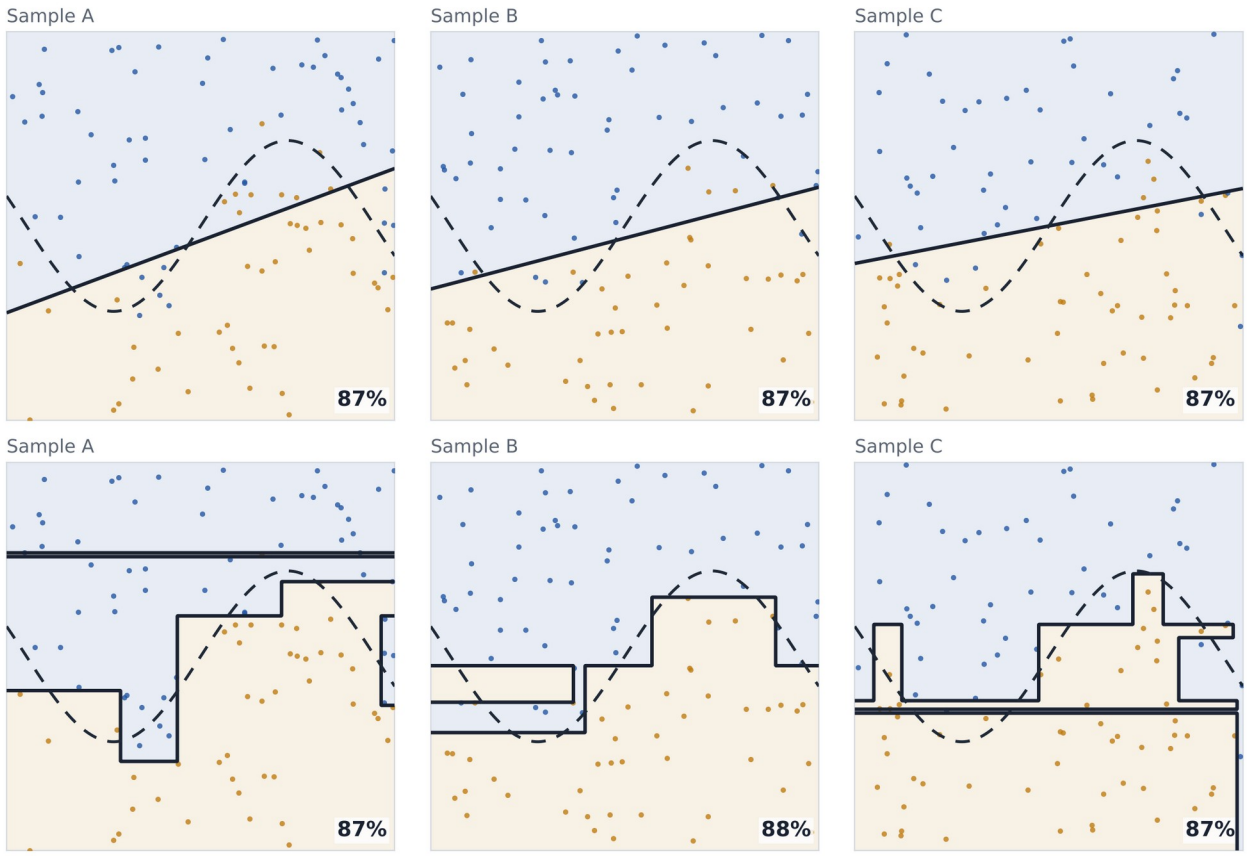
Compress features, keep structure

Many survey items to a few factors

Same question first, every time: what action follows the prediction, and when is it needed?

Bias and variance are different ways to be wrong

The same two models refit on three samples of 100 customers from one population. Dashed curve: the true boundary.



HIGH BIAS

Logistic regression

The three lines nearly coincide, and all three cut straight through the curve. More data will not fix this: a line cannot bend.

HIGH VARIANCE

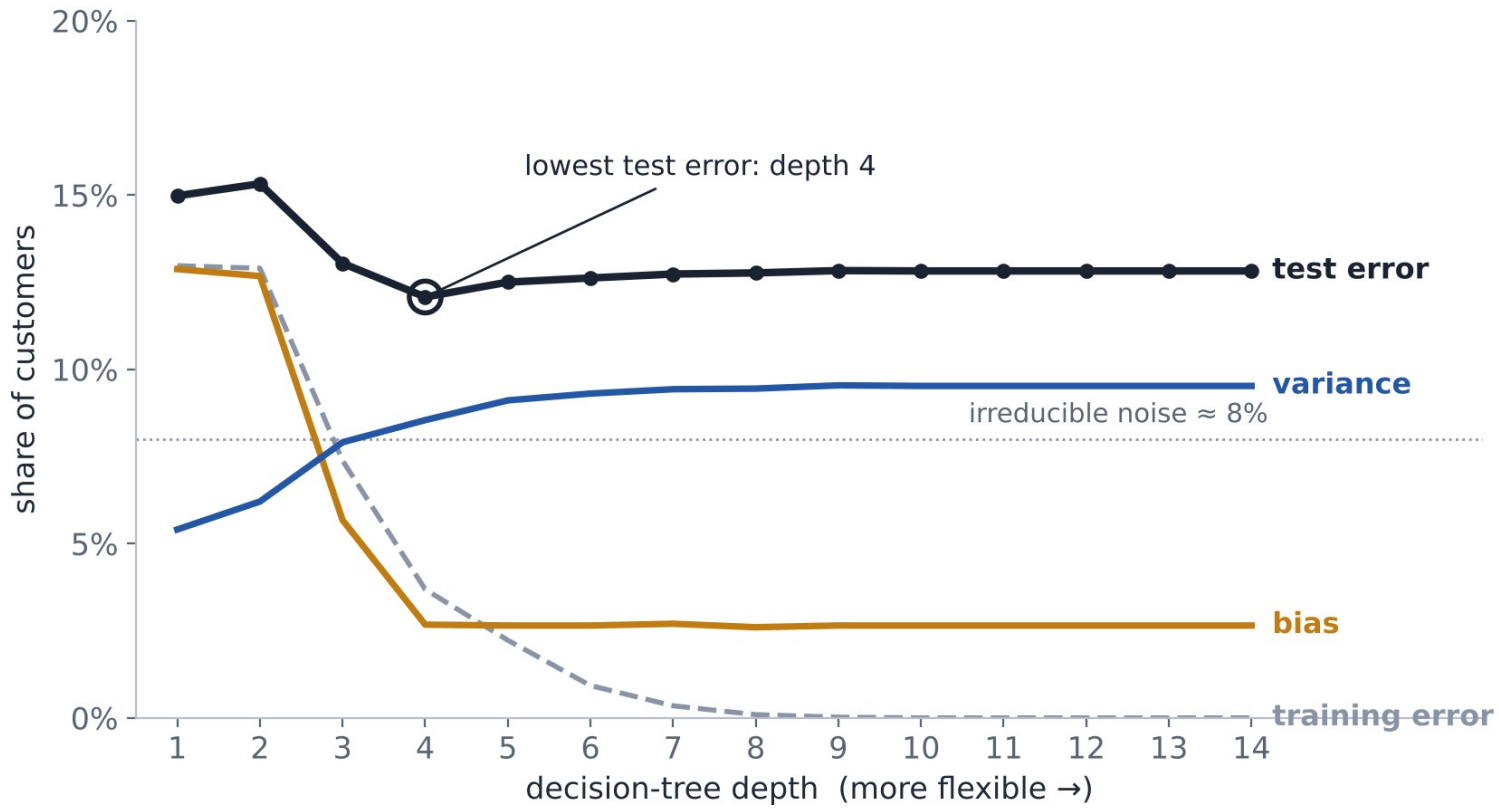
Unpruned decision tree

Every sample produces a different rule, including thin strips around single customers. The rule describes the sample, not the population.

Same accuracy on new customers (about 87%), opposite reasons. Bias asks whether the model can represent the pattern; variance asks whether the learned pattern is stable.

Model flexibility trades bias for variance

Decision trees of depth 1–14, each refit on 100 samples of 150 customers. Errors measured on 4,000 new customers.



BIAS

Average-vote prediction wrong vs. the true boundary.

Falls from 13% to 3% by depth 4, then stays flat.

VARIANCE

Refits that disagree with the average vote.

Rises from 5% to 10%: deeper trees chase each sample.

TRAINING ERROR

Falls to 0% and keeps looking better.

It cannot see variance, so it cannot pick the depth.

CHOOSE

Depth 4: the lowest error on new customers.

Found with validation data, never with training fit.

Which methods fix which failure

Every method from Weeks 1–7 either adds flexibility or constrains and averages it.

HIGH BIAS → ADD FLEXIBILITY

Symptom: training and validation error both high, and close together

- Nonlinear features and interactions
- Deeper trees; more boosting rounds
- A weaker regularization penalty

HIGH VARIANCE → CONSTRAIN OR AVERAGE

Symptom: low training error, much higher validation error

- A stronger regularization penalty
- Shallower trees, larger minimum leaf
- Bagging and random forests
- Early stopping; more training data

Training error alone cannot tell these apart. Always compare it with validation error.

EXPLORE IT MLU-Explain: The Bias–Variance Tradeoff (interactive)

Five families, five learning behaviors

Week 7's map, unchanged. Read the middle column as how each model spends its effort.

MODEL FAMILY	HOW IT LEARNS	PRIMARY EFFECT
Logistic	One global linear score in transformed feature space	Low variance; explicit coefficients
One tree	One recursive partition	Readable; unstable
Bagging	Independent trees on resampled rows, then average	Reduces variance
Random forest	Bagging plus feature sampling	Decorrelates trees
Boosting	Sequential shallow trees fit the current loss signal	Reduces remaining systematic error

Boosting is the only family here that changes its target after every stage.

The question Week 7 left open

WHAT WEEK 7 ESTABLISHED

LOSS	Every model minimizes a loss; the loss decides what counts as an error.
HARD CASES	Boosting asks “what is still wrong?”: large residuals pull the next tree hardest.
CAPACITY	Flexibility knobs are chosen on validation evidence, never on training fit.

TODAY'S QUESTION

Several boundaries classify last month's customers perfectly.

Which one should we trust with next month's?

Part 2 answers it with a new model, then we compare it with everything above.

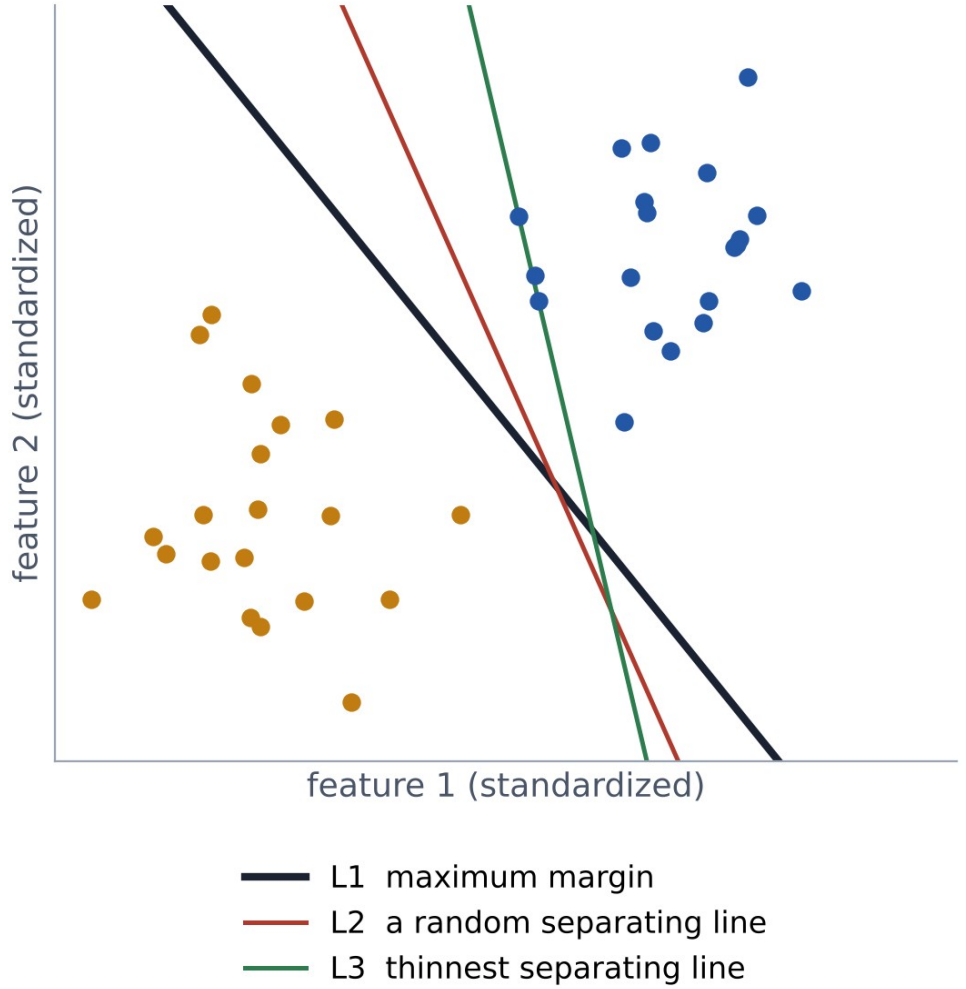
WEEK 8 · PART 2

Support vector machines: from why to how

Motivation → margin → soft margin → scaling → kernels → probabilities

The problem: several boundaries fit the past equally well

SITUATION	A classifier must sort next month's customers, not last month's.
THE TRAP	Every line on the right has 0% training error, so the leaderboard cannot choose.
THE BET	Prefer the boundary that stays farthest from every training case, so small shifts in new data do not flip decisions.
THE TEST	Only validation on new data shows whether the bet paid off.



Across 200 fresh samples, the widest margin generalizes best on average

Error on 20,000 new customers, averaging over 189 training samples of 40 customers where competing lines existed.

1.8%

Maximum margin

2.4%

Any separating line

3.2%

Thinnest separating line

The widest margin beat or tied a random separating line in 69% of samples: a sound bet, not a guarantee. That is why we still validate.

Where SVMs earn their place, and where they do not

STRONG FIT

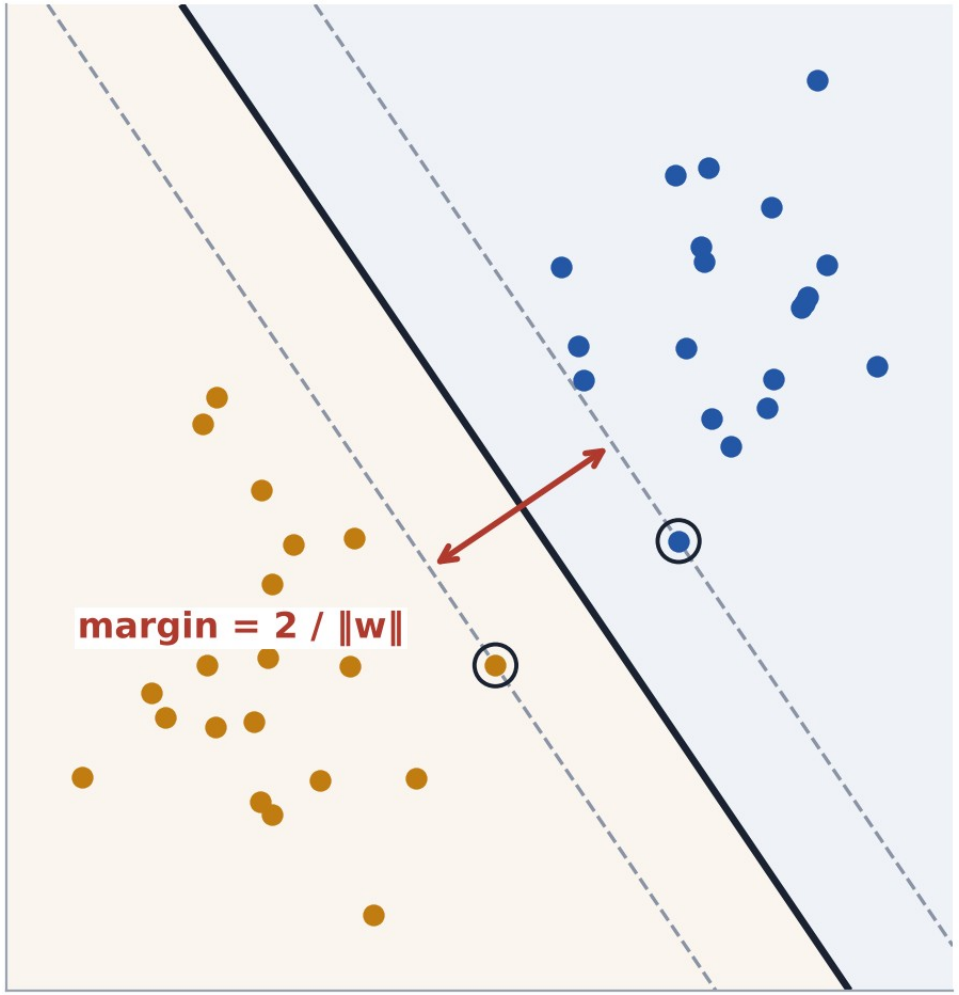
- Text on sparse word features: spam filters, support-ticket routing, sentiment
- Wide data with more features than rows: gene expression, sensor signatures
- Small-to-medium datasets with smooth nonlinear boundaries
- A robust benchmark beside regularized logistic regression

WEAKER FIT

- Kernel SVM on hundreds of thousands of rows: fit time grows at least quadratically with rows
- Mixed-type business tables, where boosting usually leads
- Decisions that need explainable coefficients or native probabilities
- Many categorical features with missing values: more preprocessing to get right

Know the tool's home ground before you put it on a leaderboard.

Picture the widest street between two neighborhoods



BOUNDARY	The center line, w·x + b = 0. The sign of the score gives the class.
MARGIN	The street's width, 2 / ‖w‖. A wider street means a smaller ‖w‖.
SUPPORT VECTORS	The houses on the curb. They alone fix where the street runs.
EVERYONE ELSE	Houses set back from the street can move or vanish; the street does not change.

So what? Borderline cases control the fit: their data quality and scaling matter most.

The whole idea fits in one optimization

$$\text{maximize } 2 / \|w\| \iff \text{minimize } \tfrac{1}{2}\|w\|^2$$

$$\text{subject to } y_i (w \cdot x_i + b) \geq 1 \text{ for every training point } i$$

SCORE

f(x) = w·x + b. The sign gives the class; the size is a distance from the boundary.

CONSTRAINT

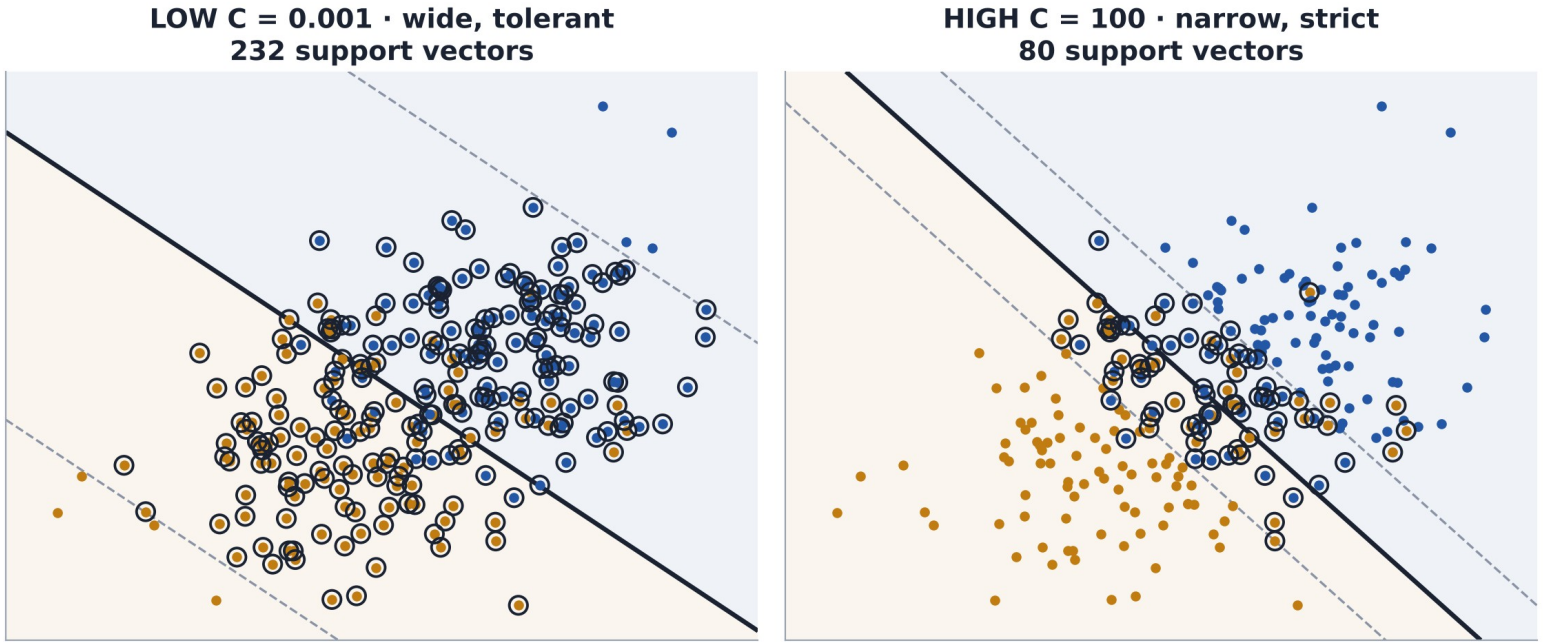
Labels are coded y = ±1. Every point must sit on its own side, at least at the margin edge.

THE CATCH

When classes overlap, no such street exists. Real data needs a softer rule: next slide.

Soft margins buy a wider street by allowing some violations

C is the price of each violation. Same 240 training customers, two prices.



C

Penalty per point inside the margin or misclassified

HIGH C

Narrow street, few violations, fewer support vectors: more variance

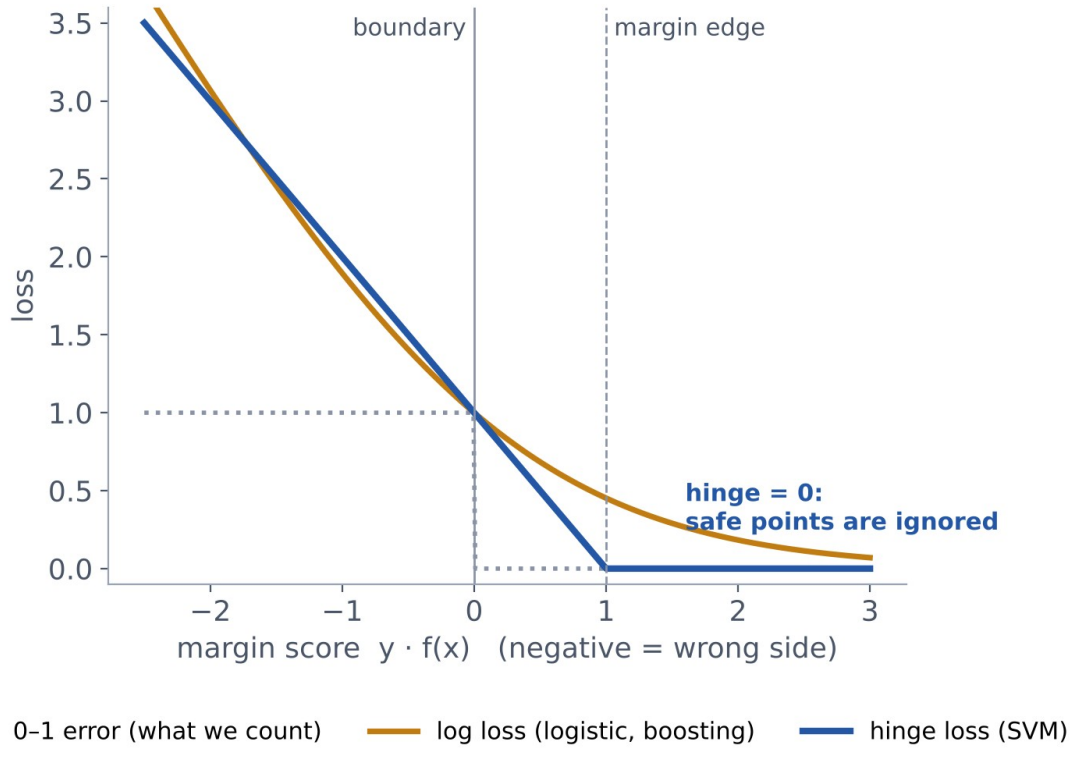
LOW C

Wide street, many support vectors: more bias, more stability

CHOOSE

By validated decision value, never by training accuracy

The SVM is Week 7's recipe with a different loss

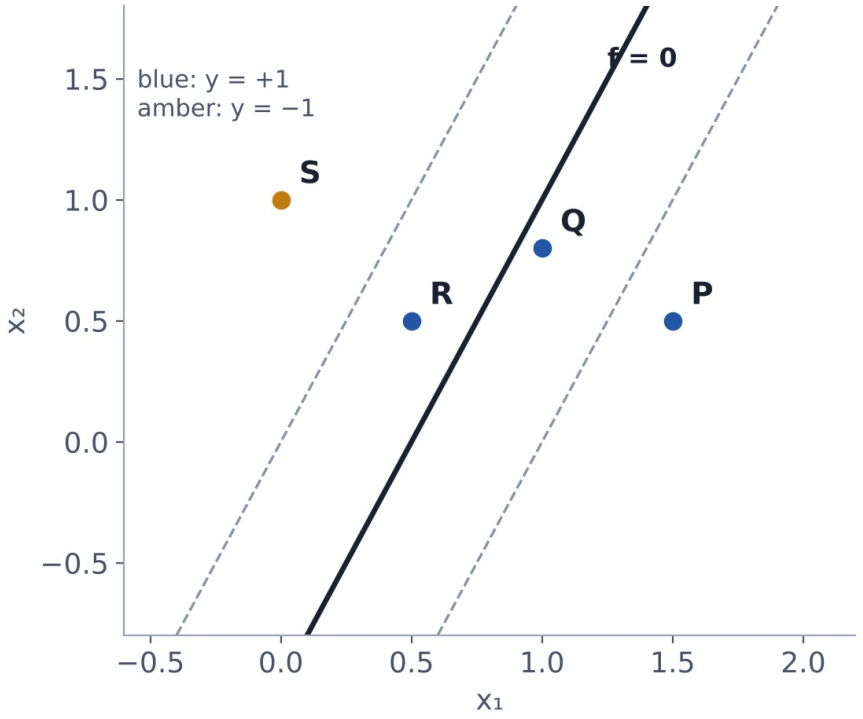


$$\text{minimize } \tfrac{1}{2}\|w\|^2 + C \cdot \Sigma \max(0, 1 - y_i f(x_i))$$

PENALTY	½‖w‖² keeps the street wide, like the logistic penalty.
LOSS	Hinge charges only points inside the margin or on the wrong side.
CONSEQUENCE	Points with y·f(x) ≥ 1 cost nothing, so only support vectors shape the fit.
CONTRAST	Log loss never reaches zero, so every row nudges logistic and boosting.

Score four customers by hand

w = (2, −1), b = −1, so $f(x) = 2x_1 - x_2 - 1$. Margin width = 2 / ‖w‖ = 2 / √5 ≈ 0.89



CUSTOMER	X	Y	F(X)	Y·F(X)	HINGE	STATUS
P	(1.5, 0.5)	+1	1.5	1.5	0	Safe
Q	(1.0, 0.8)	+1	0.2	0.2	0.8	Inside margin
R	(0.5, 0.5)	+1	−0.5	−0.5	1.5	Misclassified
S	(0.0, 1.0)	−1	−2.0	2.0	0	Safe

If this were the fitted model, Q and R would be support vectors. P and S contribute nothing to the loss.

Commit to a decision before seeing the model answer

A churn SVM uses annual spend in dollars and app sessions per week (0–20). Spend dominates the geometry.

A

Interpret as-is

B

Scale then refit

C

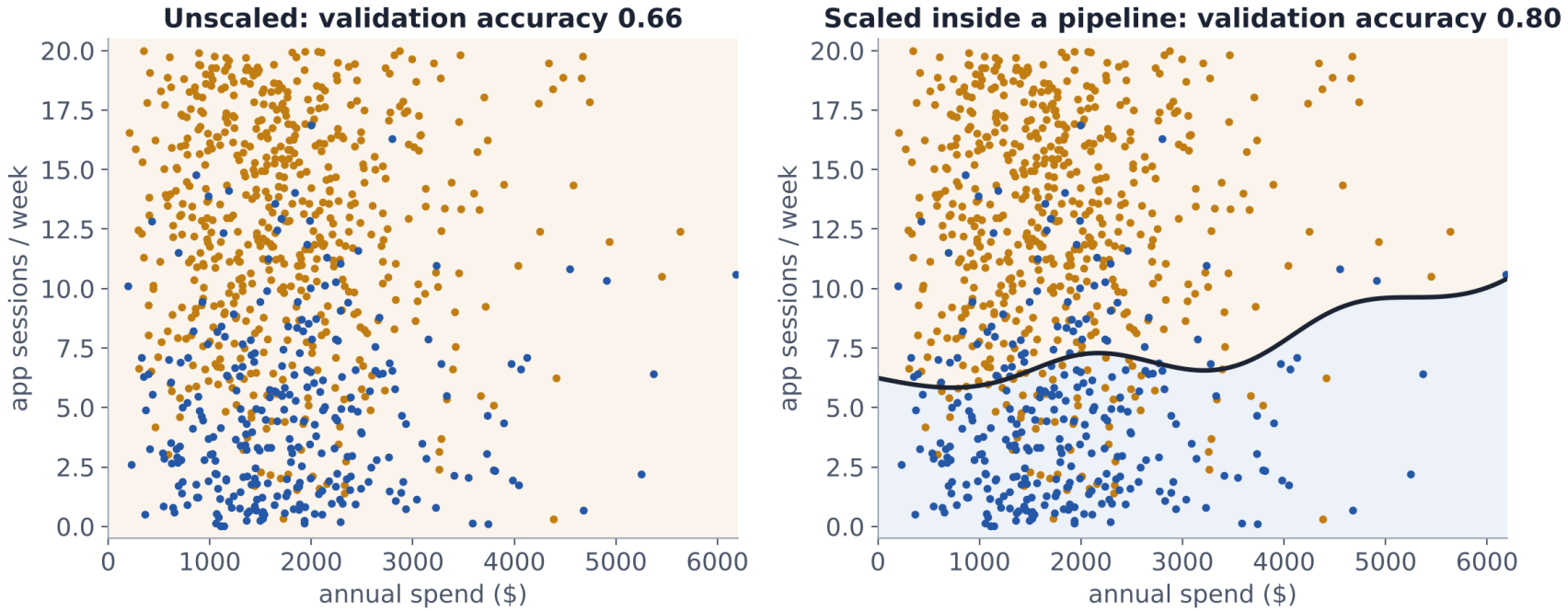
Raise C only

Vote silently → defend with one piece of evidence → name what would change your mind.

Distance-based geometry is meaningful only after feature scales are defensible.

Scaling changes distance, and therefore the model

Same RBF SVM, same 840 training customers. Churn (blue) is driven mainly by low app sessions.



UNITS

Unscaled, dollars swamp sessions: 0.66 accuracy, exactly the majority baseline.

STANDARDIZE

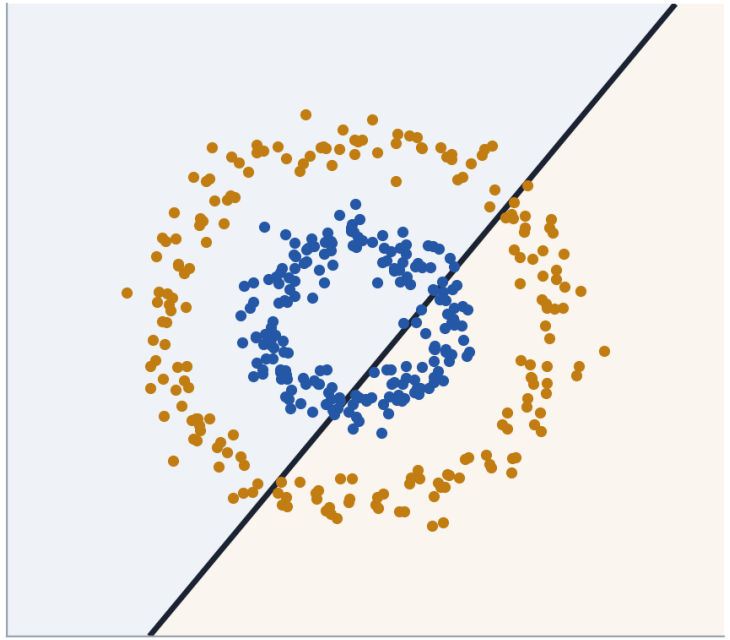
Mean 0, standard deviation 1 for every numeric feature: 0.80 accuracy.

PIPELINE

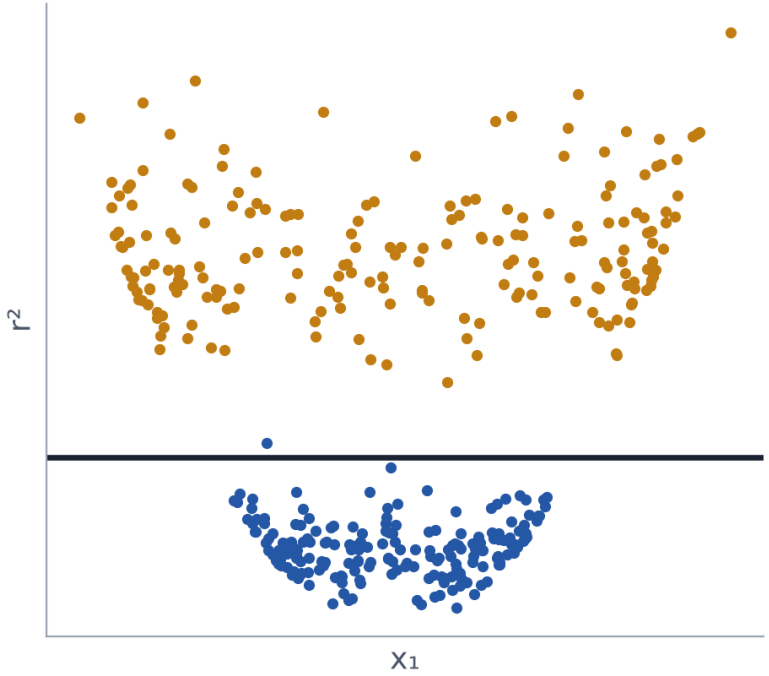
Fit the scaler on training rows only, inside every fold. Scaling all rows first is leakage.

When no line works, add features, or let a kernel add them for you

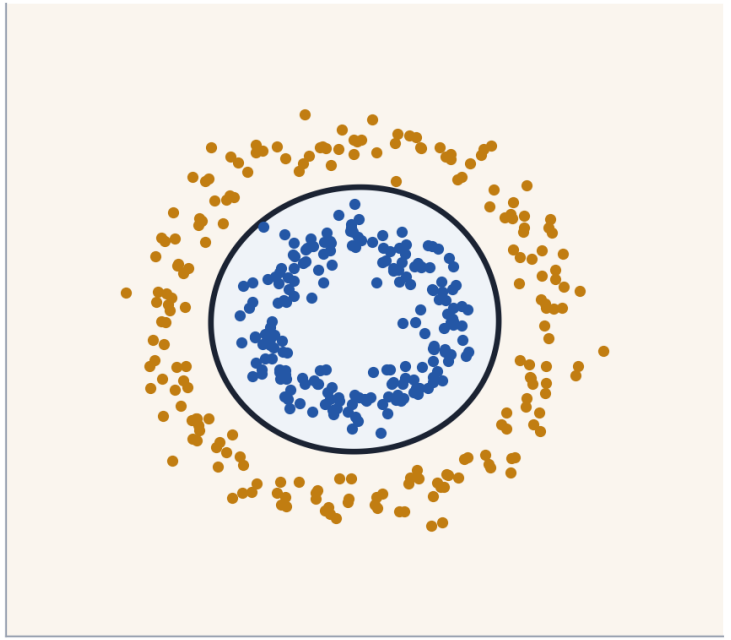
1 · Linear: train acc 0.56



2 · Add feature $r^2 = x_1^2 + x_2^2$



3 · RBF kernel: train acc 1.00



1 · LINEAR

Rings cannot be split by a line: 0.56 training accuracy.

2 · ADD A FEATURE

With $r^2 = x_1^2 + x_2^2$, a flat cut separates them: the move you already know.

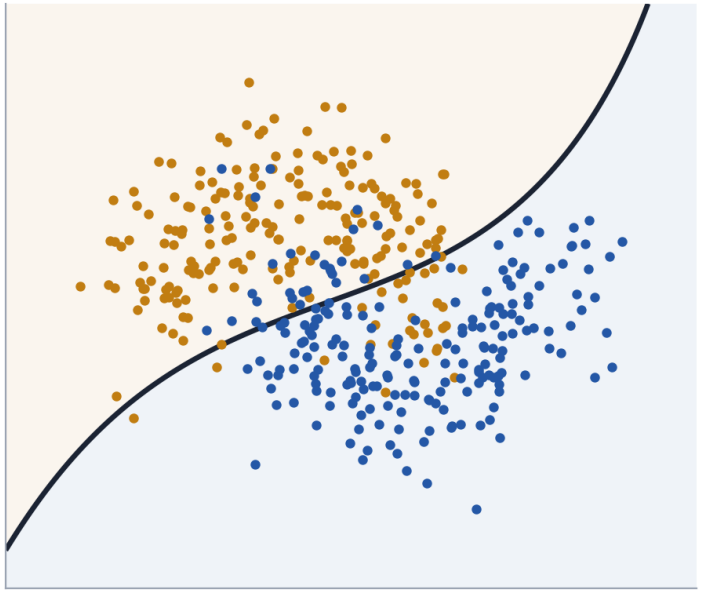
3 · RBF KERNEL

Same result without building the feature: similarity decays with distance.

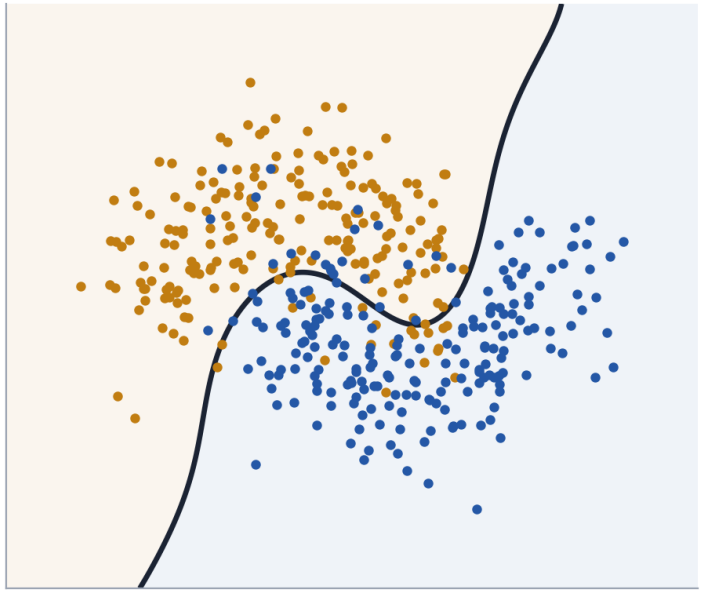
Gamma sets each point's reach: the bias–variance dial for kernels

Same 420 training rows, C = 1. Holdout = 180 rows the model never saw.

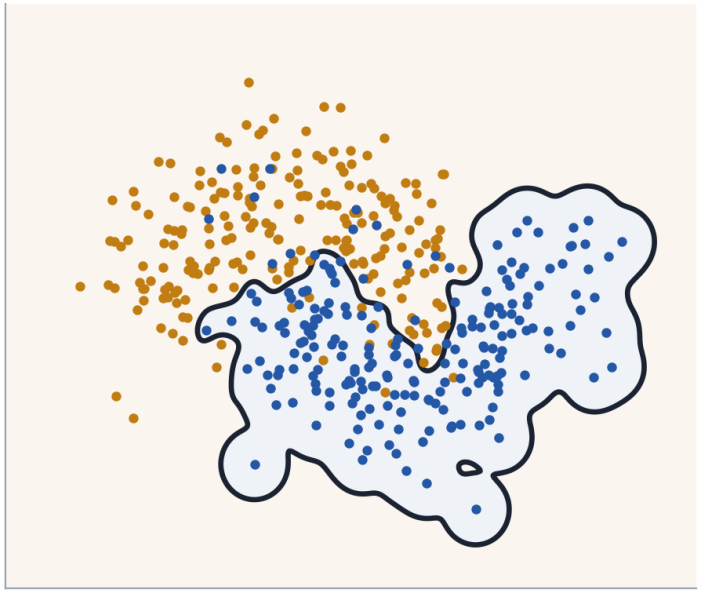
γ = 0.1 · too smooth
train 0.86 · holdout 0.82



γ = 1 · balanced
train 0.90 · holdout 0.89



γ = 50 · memorizes
train 0.94 · holdout 0.89



γ = 0.1

Train 0.86 · holdout 0.82. Too smooth: high bias.

γ = 1

Train 0.90 · holdout 0.89. Balanced.

γ = 50

Train 0.94 · holdout 0.89. Memorizes islands: high variance.

Tune C and gamma together, as you tuned learning rate and rounds

5-fold cross-validated accuracy on training rows only (workbook Task 4).

	γ = 0.1	γ = 1	γ = 10	γ = 50
C = 0.1	0.860	0.883	**0.907**	0.855
C = 1	0.860	0.893	**0.907**	0.871
C = 10	0.869	0.905	0.881	0.836
C = 100	0.893	0.905	0.819	0.831

PAIR More reach × a stricter penalty = more capacity. They interact.

SEARCH Log-spaced grid, folds built from training rows only; time-aware folds when time matters.

SELECT The simplest setting within noise of the best: here C = 0.1, γ = 10.

TEST ONCE 0.878 on untouched rows. No more tuning after this number.

An SVM returns a distance, not a probability

DECISION_FUNCTION	Signed distance from the boundary. On the workbook churn data it ranges from −1.25 to 0.59. Not a probability.
PROBABILITY=TRUE	Platt scaling with internal 5-fold cross-validation. Slower; probabilities can disagree with predict().
CALIBRATEDCLASSIFIERCV	An explicit score → probability mapping you can inspect. Check it with a reliability curve, log loss and Brier score on later data.

WHY IT MATTERS FOR BOOKINGGUARD

The break-even deposit threshold in the worked example is a probability: 0.127.

A threshold applied to an uncalibrated score has no economic meaning, even when the ranking is excellent.

A five-step workflow for any SVM

1 Pipeline

Scaler + SVC together, so preprocessing refits inside every fold

2 Start linear

Move to RBF only if validation improves materially

3 Grid C (and γ)

Log scale, time-aware folds, training rows only

4 Calibrate

Map scores to probabilities on a later window

5 Test once

Refit the selected pipeline; evaluate on untouched data

```
pipe   = make_pipeline(StandardScaler(), SVC(kernel="rbf"))
grid   = {"svc__C": [0.1, 1, 10, 100], "svc__gamma": [0.01, 0.1, 1]}
search = GridSearchCV(pipe, grid, cv=TimeSeriesSplit(5), scoring="roc_auc").fit(X_train, y_train)
cal    = CalibratedClassifierCV(FrozenEstimator(search.best_estimator_), method="sigmoid")
cal.fit(X_calibration, y_calibration)          # a later window than X_train
p      = cal.predict_proba(X_validation)[:, 1]  # now a threshold has economic meaning
```

Requires scikit-learn ≥ 1.6 for FrozenEstimator. Rows sorted by time for TimeSeriesSplit.

SVM Margin Explorer: move C and gamma, watch the margin respond



DATASET

Separable | Overlap | Rings | Moons | Spend vs sessions | Blank

CLICKING THE PLOT WILL

Add +1 | Add −1 | Remove point

Shift-click adds the other class.

MODEL

Linear kernel | RBF kernel

C · price of each violation 1

γ · reach of each point 1

RBF kernel: similarity decays with distance at rate γ.

FITTED MODEL

64/200	91.5%	50.0%
support vectors	training accuracy	majority-class baseline

Fitted 200 points in 29 ms.

OPEN THE EXPLORER

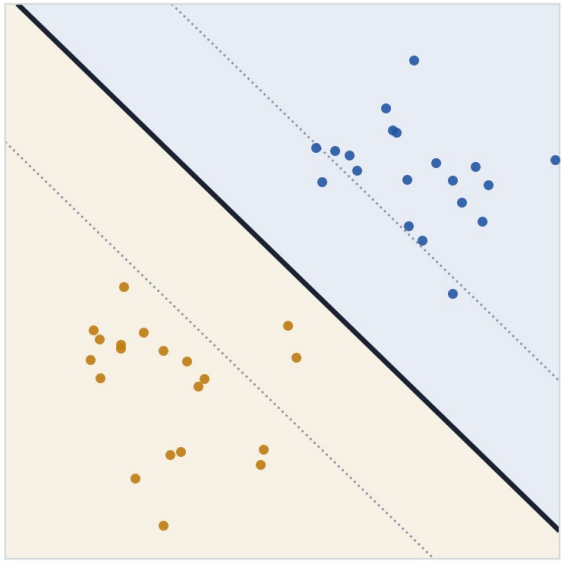
SVM Margin Explorer

Separable	Remove a far point, then a ringed one. Only the second moves the line.
Overlap	Slide C from 0.001 to 1000. The street narrows; support vectors fall.
Spend vs sessions	Untick Standardize. Dollars swamp distance.
Rings	Compare the linear and RBF kernels.
Moons	Move γ from 0.01 to 100. Training accuracy rises; does the boundary?

SVM adds an explicit margin and a controlled path to nonlinearity

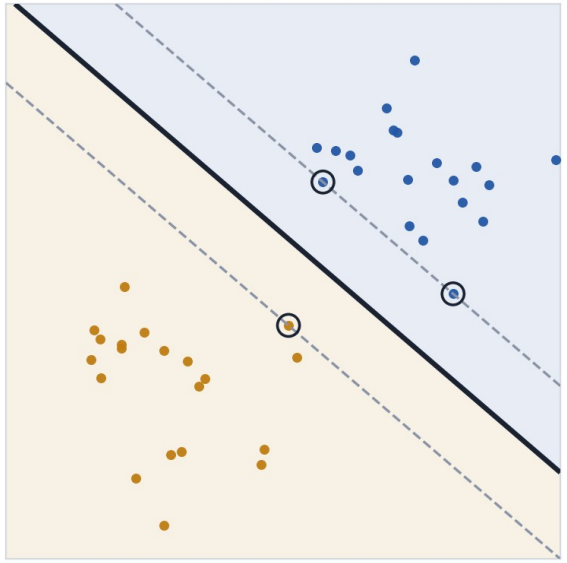
Four real fits. Solid line: boundary. Dashed: margin edges (panel 1 dotted: 10% and 90% probability). Rings: support vectors.

1 · Logistic regression



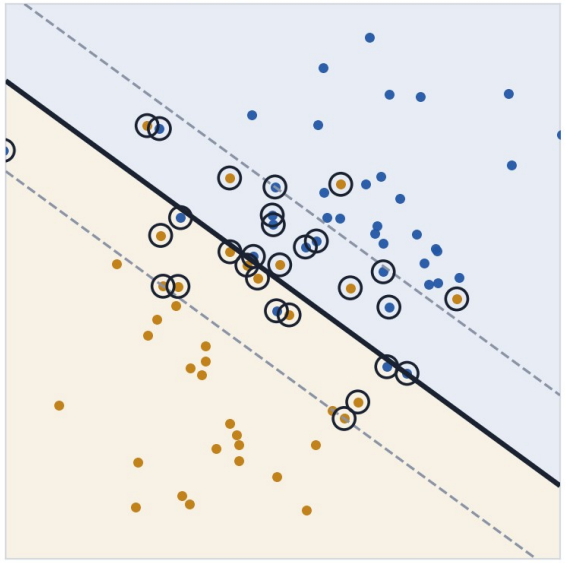
Every customer pulls on the line through log loss.

2 · Maximum margin



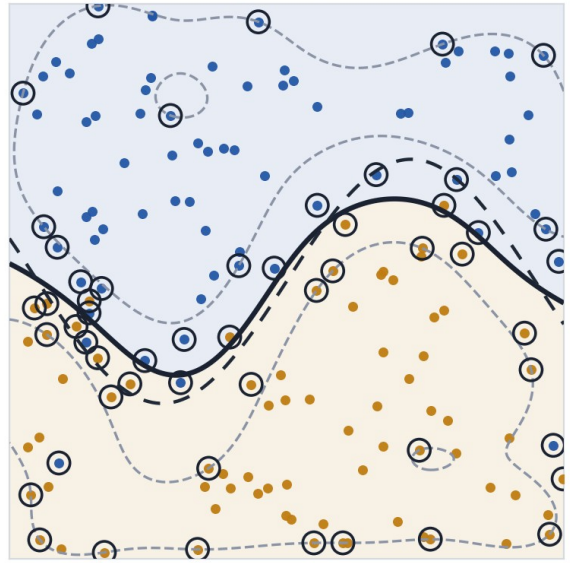
Widest street between classes. Only 3 customers fix it.

3 · Soft margin (C = 1)



C prices points inside the street. 29 of 80 now matter.

4 · RBF kernel (γ = 1)

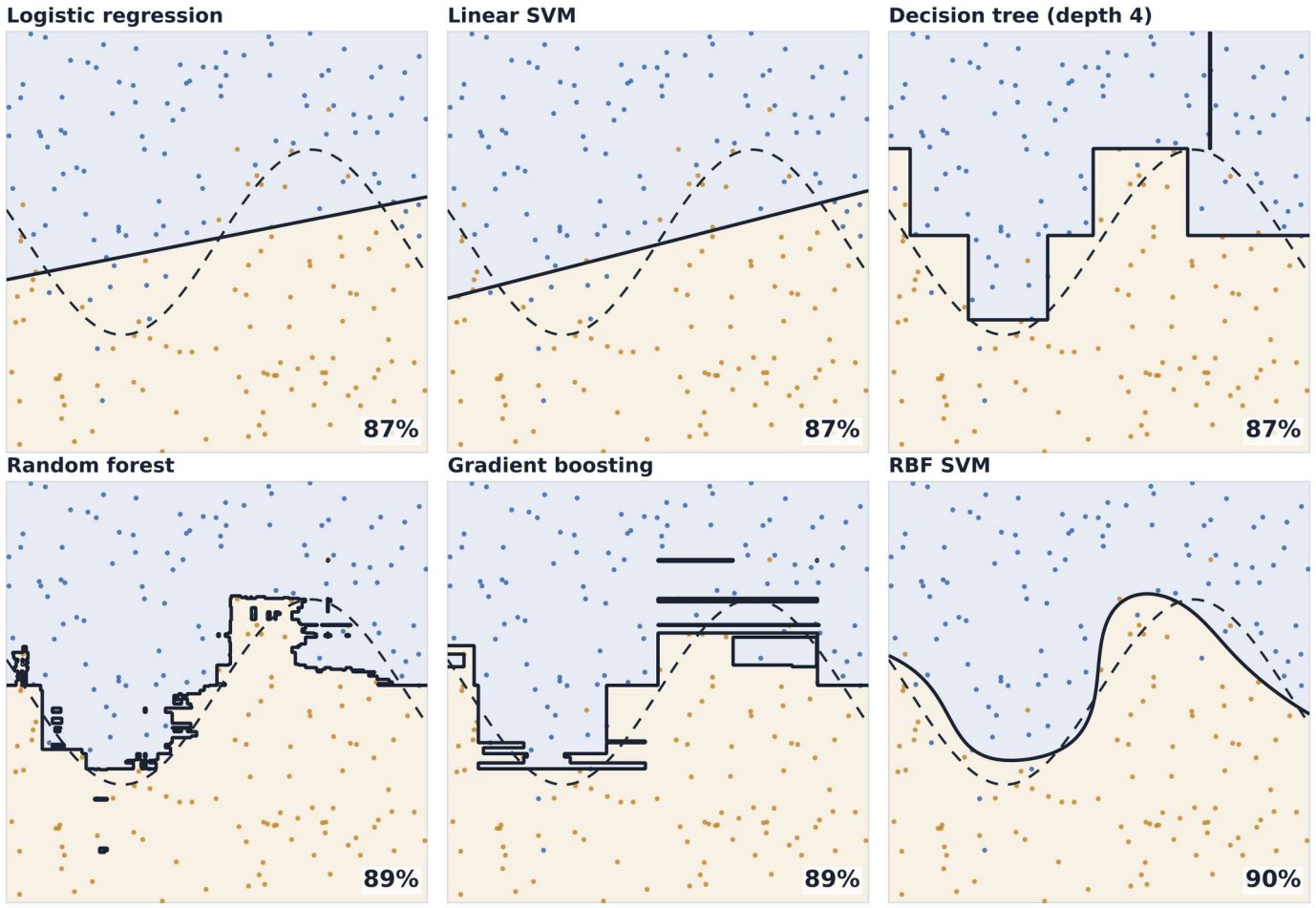


Similarity to support vectors bends the boundary. γ sets how local.

DISCIPLINE Standardize features, tune C and γ on validation data, evaluate the test set once.

Six ways to turn customer features into a classification rule

One sample of 200 customers, six models. Big number: accuracy on 20,000 new customers. Dashed curve: the true boundary.

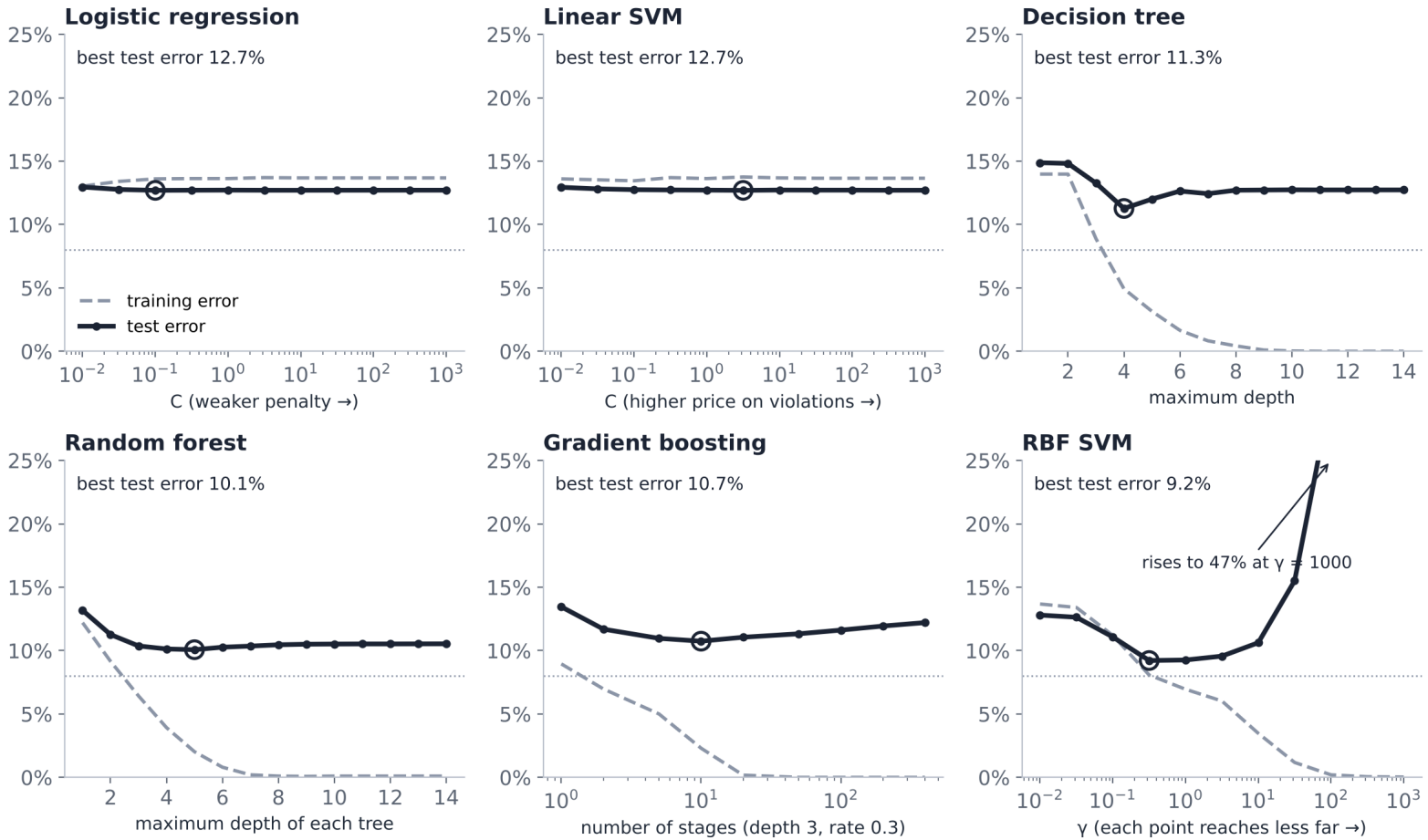


MODEL	LEARNS	BOUNDARY
Logistic	linear log-odds	straight line
Linear SVM	widest-margin line	straight line
Tree	if/then splits	axis-aligned steps
Random forest	average of many trees	smoother steps
Boosting	sum of small trees	refined steps
RBF SVM	similarity to support vectors	smooth curve

Only the flexible families can follow the curve. Whether they follow it stably is the next question.

Every model family has a flexibility control

Training and test error as each family's knob moves. Averages over 20 samples of 200 customers; dotted line = 8% irreducible error.



LINEAR

Flat curves stuck at 12.7%.

The knob cannot add curvature: bias stays.

TREES

A shallow U, best near depth 4–5.

Averaging (forest) flattens the right side.

RBF SVM

Lowest error here: 9.2%.

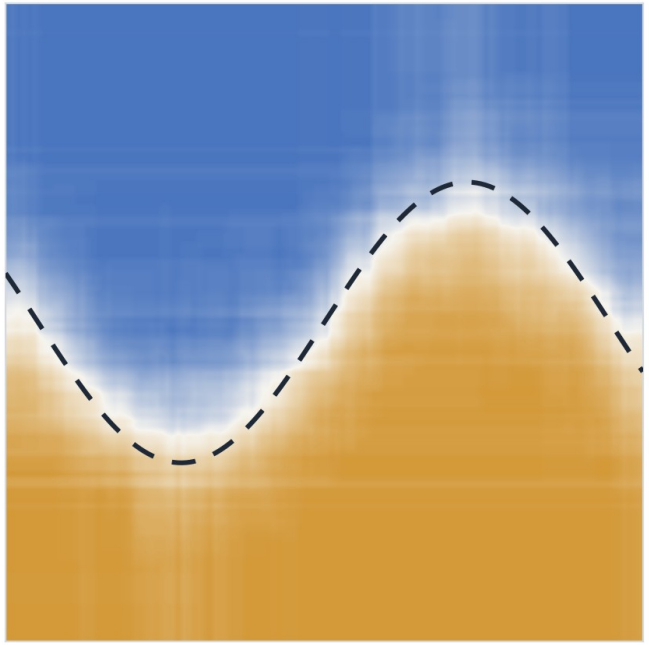
Push γ too far and test error climbs to 47%.

The algorithm name does not fix bias or variance. The knob setting does.

Averaging and margins reduce boundary movement

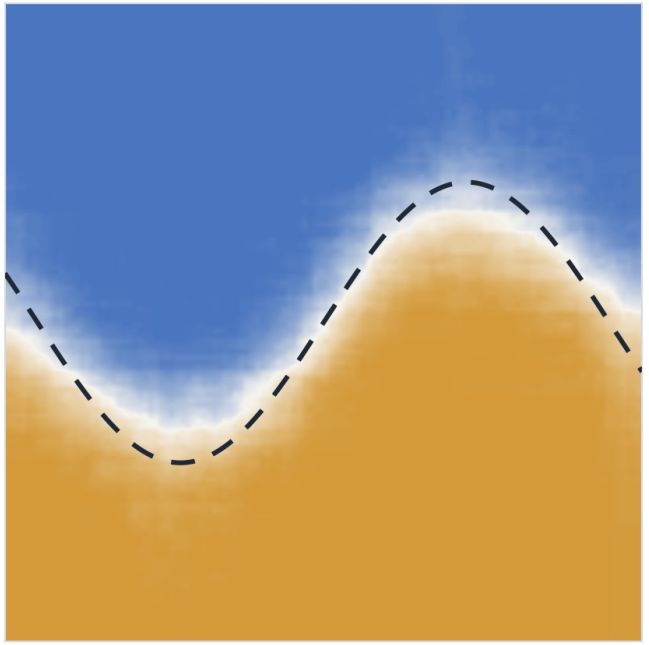
Each map stacks 100 refits on 150-customer samples. Color = share of refits predicting blue; white = the refits disagree.

Unpruned decision tree



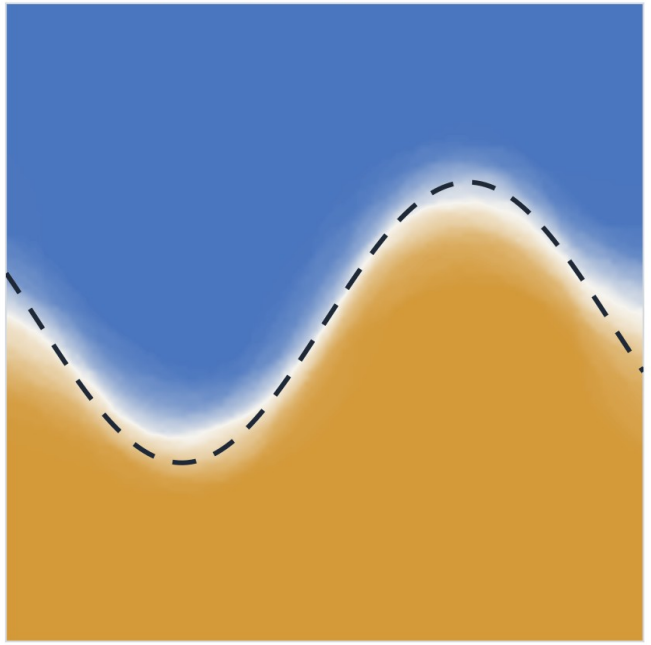
Test error 12.9%
bias 2.9% · variance 9.6%

Random forest (averages 200 trees)



Test error 10.5%
bias 3.5% · variance 6.3%

RBF SVM (C = 1, γ = 1)



Test error 9.4%
bias 2.4% · variance 4.7%

■ all refits say blue ■ refits split ■ all say amber

Forests cut variance by averaging trees. The RBF SVM gets a smooth, stable boundary from its margin.

Every building block returned in the SVM

The recap from Part 1, now with the links named.

BUILDING BLOCK	WHAT IT GAVE US	WHERE IT SHOWED UP TODAY
Probability & thresholds	Scores become actions through costs	SVM scores must be calibrated before a threshold
Validation	Evidence that represents future use	C and gamma are chosen on validation folds
Regularization	A penalty that trades fit for stability	$\frac{1}{2}\|w\|^2$ is the penalty; C sets its price
Nonlinear features	Curved boundaries by adding features	Kernels add those features implicitly
Trees	Piecewise rules; feature scale does not matter	Contrast: SVM geometry depends on scaling
Forests	Averaging cuts variance	Contrast: SVM gets stability from the margin
Boosting	Loss + gradient steps + capacity control	Same recipe, different loss: hinge

From “what is still wrong?” to “where is the safest line?”

	GRADIENT BOOSTING · WEEK 7	SUPPORT VECTOR MACHINE · TODAY
Core question	What does the ensemble still get wrong?	Where is the widest safe street between classes?
Hard cases get attention by	Large residuals pull the next tree hardest	Only points on or inside the margin shape the fit
Loss	Log loss (classification)	Hinge loss
Capacity controls	Learning rate × rounds, depth, early stopping	C, kernel, gamma, chosen by validation
Preprocessing	Trees ignore feature scale	Standardize every feature, inside the pipeline
Output	Log-odds score that maps to a probability	Signed distance; calibrate before thresholds

Both focus on hard cases. Boosting does it by reweighting errors; the SVM does it by geometry.

Every classifier this term answers the same five questions

Read across a row to compare models; read down a column to describe one.

QUESTION	LOGISTIC	RANDOM FOREST	BOOSTING	SVM
Minimizes what?	Log loss + penalty	Impurity, split by split; then averages	Log loss, one tree per gradient step	Hinge loss + ½‖w‖²
Capacity knob?	Penalty, features	Leaf size, features per split	Learning rate × rounds, depth	C, kernel, gamma
Needs from preprocessing?	Scaling for the penalty; encoding	Encoding only	Encoding only	Standardization is essential
More flexible (less bias)	More features, weaker penalty	Deeper trees, smaller leaves	More rounds, deeper trees	Higher C, larger γ
More stable (less variance)	Stronger penalty	More trees averaged	Early stopping, smaller rate	Lower C, smaller γ: a wider margin
Outputs?	Probability	Vote share; check calibration	Probability; check calibration	**Distance; calibrate first**

The last row is why calibration matters for BookingGuard: the deposit threshold is a probability.

Which classifier to try first

Week 7's decision tree, with the SVM added. Every path ends at the same validation gate.

1	Must you explain each factor's effect, or need native probabilities?	yes →	**LOGISTIC FIRST** Credit scorecards, insurance pricing
2	Sparse text, or more features than rows?	yes →	**LINEAR SVM** Ticket routing, spam, document tagging
3	Up to tens of thousands of rows with a smooth nonlinear boundary?	yes →	**RBF SVM** Sensor fault detection, small clinical or survey data
4	Large mixed-type business table?	yes →	**BOOSTING** Churn, cancellations, demand: HistGB or LightGBM

Every candidate then faces the same rows, time split, metrics, calibration check and policy constraints.

WEEK 8 · PART 3

In-class lab: the SVM workbook

Predict, change, run, record, decide

Instructor demonstration, then paired investigation

Open Blackboard → Week 8 → In-class SVM workbook. Save a copy in Drive before editing.

8 MIN · WATCH	Instructor runs Part A (why a margin) and Part B (which points decide the boundary).
20 MIN · INVESTIGATE	Pairs complete Tasks 1–5: predict in the markdown cell first, then run and record.
5 MIN · DEBRIEF	Pause after Task 4 and Task 5: what changed, which evidence, what would change the decision.

WORKBOOK PART	QUESTION IT ANSWERS	MINUTES
Part A · Part B	Why prefer one separating line? Which points decide it?	8
Task 1	Hinge loss by hand	4
Task 2	What does C trade off?	4
Task 3	Why must features be scaled?	3
Task 4	What do kernels and gamma buy, and at what risk?	4
Task 5	Can an SVM score support a threshold policy?	5

BLACKBOARD › Week 8 › In-class SVM workbook

Task 1 · Hinge loss by hand

w = (1.5, −1), b = −0.5. Work in pairs. Show the arithmetic, then one sentence of interpretation.

CUSTOMER	X_1	X_2	Y	F(X)	Y·F(X)	HINGE	STATUS
A	2.0	1.0	+1				
B	1.0	0.5	+1				
C	0.5	1.0	−1				
D	1.0	−1.0	−1				

DELIVERABLE

Fill the four columns, compute the margin width 2/‖w‖, then explain in one sentence why customer A would not change the fitted boundary.

BLACKBOARD › Week 8 › In-class SVM workbook › Task 1

Task 2 · What does C trade off?

Linear SVM on overlapping classes. Change C_VALUES in the cell; record one row per value. ✏

C	PREDICTED DIRECTION	SUPPORT VECTORS	MARGIN WIDTH	TRAIN ACC	VAL ACC	KEEP / REJECT
0.001						
0.1						
10						
100						

PREDICT

As C rises, which way do support vectors, margin width, training accuracy and validation accuracy move?

EXPLAIN

In one sentence: why is C the SVM's version of the regularization penalty from logistic regression?

BLACKBOARD › Week 8 › In-class SVM workbook › Task 2

Task 3 · Why must features be scaled?

RBF SVM, annual spend ($) and app sessions per week. Compare unscaled with a StandardScaler pipeline.

VERSION	YOUR PREDICTION	VALIDATION ACCURACY	MAJORITY BASELINE	WHAT DOMINATED DISTANCE?
Unscaled				
Scaled in a pipeline				

COMPARE

Is the unscaled model better than predicting the majority class for everyone? What does that tell you?

TRANSFER

Would a random forest on the same two features need the scaler? Defend your answer in one sentence.

BLACKBOARD › Week 8 › In-class SVM workbook › Task 3

Task 4 · Kernels and gamma: flexibility, and its price

RBF SVM on curved data. Read the three panels, run the C × γ grid, then test once.

RECORD	YOUR VALUE
Which γ overfits? Evidence from the three panels	
Best (C, γ) from the 5-fold grid	
Best CV accuracy	
Test accuracy, run once	
Gap between CV and test, and why it exists	

EXPLAIN · WEEK 7 LINK

Why are C and γ one decision, not two, in the same way that learning rate and number of rounds were one decision for boosting?

BLACKBOARD › Week 8 › In-class SVM workbook › Task 4

Task 5 · From an SVM score to a policy decision

Simulated churn. Train months 1–16, validate 17–20, test 21–24. Offer costs $20 and recovers $80, so break-even p = 0.25.

MODEL	LOG LOSS	PR-AUC	BRIER	CONTACTED @ 0.25	NET VALUE	FIT SECS
Logistic						
SVM (RBF) + calibration						
HistGB						

FOUR LINES (SAME AS WEEK 7)

- Model and exact configuration
- Two pieces of evidence
- One risk
- The next check before deployment

SENSITIVITY

Set COST = 30 and rerun. Which moves: the model ranking, the threshold, or both? Then run the test cell once, for your chosen model only.

BLACKBOARD › Week 8 › In-class SVM workbook › Task 5

Retrieval check

Complete individually before the group debrief.

1. Several lines separate the training data perfectly. What property does the SVM use to choose, and why should it help next month?

2. A point lies far outside the margin on the correct side. What is its hinge loss, and what happens if you delete it?

3. An RBF SVM reaches the highest training accuracy at $\gamma = 50$ but loses to $\gamma = 1$ on validation. Name the error pattern and the evidence.

4. Why can't you set a deposit or retention threshold directly on decision_function output? What do you do instead?

BLACKBOARD › Week 8 › Retrieval check

WEEK 8 · PART 4

Model selection: from a leaderboard to a decision

You now have five classifier families. Which one goes into production is a separate question.

Model selection starts before algorithms

Four questions settle what 'best' means. Answer them before opening a notebook.

QUESTION	ASK	BOOKINGGUARD ANSWER
Decision	What action follows the prediction?	Require a deposit, or not
Unit	What receives one prediction?	One booking
Moment	When must the output exist, and what is known then?	A fixed point before arrival; only fields known by then
Cost	Which errors matter, and how much?	A missed cancellation loses room revenue; an unnecessary deposit adds guest friction

If two people answer these four differently, they will disagree about the 'best' model, and both will be right.

Data geometry narrows the candidate set

WHAT THE DATA LOOKS LIKE	TRY FIRST	WHY
Many rows, a few dozen mixed features	Boosting, logistic	Interactions and scale; fast to tune
Wide or sparse (text, flags)	Linear SVM, regularized logistic	Linear boundaries work well in high dimensions
Modest rows, smooth nonlinear boundary	RBF SVM, boosting	Kernels bend smoothly; training stays affordable
Rows ordered in time	Any of the above	**The validation design dominates the model choice**

BookingGuard is a time-ordered, mixed-type table: start from logistic and boosting, and let the temporal split judge.

Baselines show how hard the problem is; evidence is a portfolio

BASELINES

Rule: the existing policy (no deposits, or deposits for every non-refundable rate)

Naive: majority class, recency, or the training-period rate

Linear: logistic regression with honest preprocessing

Value: incremental decision gain over the best of the above

EVIDENCE PORTFOLIO

Primary: a metric aligned to the decision (log loss, policy value)

Calibration: do predicted risks match observed rates?

Slices: who loses: hotel, market segment, new vs repeat guest

Stability: across folds, seeds and time periods

A selection matrix makes the trade-offs explicit

Illustrative starting judgments for a mid-sized business table. Your validation evidence overrides every cell.

CRITERION	LOGISTIC	RANDOM FOREST	BOOSTING	RBF SVM
Quality	Strong when effects are near-linear	Good with interactions	Often strongest on tabular data	Strong on smooth boundaries
Robustness	Stable across samples	Stable; averaging removes variance	Sensitive to rate and rounds	Sensitive to scaling, C and γ
Operability	Fast to train and score	Large model files	Fast scoring; tuning takes time	Slow beyond tens of thousands of rows
Governance	Coefficients explain direction	Needs local explanations	Needs local explanations	Hard to explain; calibrate first

Choose the simplest model whose gain is material, stable, reproducible and operationally acceptable.

Interpretability is a property of the use case, not the model

Four questions different stakeholders will ask about the same deposit model.

GLOBAL

Revenue manager

Which factors drive deposit requests across the portfolio?

LOCAL

Front desk

Why did this guest receive a deposit request?

PROCESS

Auditor

Which data, model version, threshold and date produced it?

CONTEST

Guest or agent

How is a wrong request corrected, and how fast?

Design the explanation for the person who must act on it.

A decision memo: recommendation plus reversibility

Illustrative example: a card-fraud review queue. Use the same four headings in BookingGuard.

CHOOSE	Linear SVM with sigmoid calibration on sparse merchant and device flags; review when p ≥ 0.08.
WHY	Recall +6 points at equal review volume versus logistic on March–April validation; calibration within tolerance.
WATCH	Weekly calibration by merchant category; daily review volume above the team's capacity of 1,200.
FALLBACK	The current logistic model runs in shadow; revert if precision falls two weeks in a row.

Selection check

A model wins AUC by 0.002 but triples latency and loses calibration. What is the best default decision?

A

Ship it

B

Reject automatically

C

Quantify decision value

D

Tune on test

Commit → discuss with a neighbor → reveal. Explain the mechanism, not merely the letter.

EXERCISE 03 · LAUNCHES TODAY

BookingGuard: which bookings should require a deposit?

Individual exercise · due Sunday, November 15, 11:59 PM ET

Protect revenue without requiring a deposit from everyone

PREDICT	Cancellation risk for one booking at a fixed point before arrival.
VALIDATE	Use earlier arrivals to predict later arrivals.
COMPARE	Logistic regression, random forest and one boosting implementation.
DECIDE	A deposit threshold chosen from the published costs.
CHECK	Calibration, deposit volume, cost sensitivity and three operational groups.
SUBMIT	Executed notebook, experiment record, one-page memo, decision file, AI-use disclosure.

From a booking record to a deposit decision

Prediction is one stage of four. Each stage has its own question and its own failure mode.

1

BOOKING FACTS

Is every field known at the decision moment?

FAILS BY Leakage

2

CANCELLATION RISK

Are probabilities calibrated on later arrivals?

FAILS BY Drift, miscalibration

3

POLICY THRESHOLD

Does the threshold follow the published costs?

FAILS BY 0.50 by habit

4

DEPOSIT ACTION

Who bears the friction, and how many?

FAILS BY Volume or group harm

Validate data timing, probability calibration, cost assumptions and policy effects separately.

A feature is valid only if it exists at the decision point

AVAILABLE	Lead time, stay details, customer history, booked channel.
TARGET	canceled: appears in training data only.
EXCLUDE	reservation_status and any post-arrival or post-cancellation information.
AUDIT	Write down when each field becomes known, before fitting anything.

First checkpoint: state the prediction moment in one sentence, then list every field that fails it.

Later data is the real test

Random cross-validation mixes demand regimes and hides drift.

FIT
Earlier arrivals: fit preprocessing and models

VALIDATE
Latest labeled months before July 2017

TEST
July–August 2017: no labels

time →

FREEZE	Fit preprocessing on training rows only.
MEASURE	Log loss, PR-AUC, ROC-AUC, calibration and policy value on the later period.
DECIDE	Keep, revise or reject the model and threshold.
REPORT	What changed when the calendar moved forward.

A deposit policy begins with expected value

Worked example with the published policy costs.

1	**Booking revenue**	3 nights × $150	**$450**
2	**Cancellation loss**	0.70 × $450	**$315**
3	**Deposit recovery**	0.45 × $315	**$141.75**
4	**Break-even probability**	$18 ÷ $141.75	**≈ 0.127**

READ IT AS A RULE

Require a deposit when calibrated p(cancel) ≥ 0.127.

$p^* =$ deposit cost ÷ (recovery rate × loss rate × booking revenue)

Every input is an assumption: the sensitivity analysis varies each one.

This is why the SVM's distance score must be calibrated: 0.127 is a probability.

A probability becomes useful only through a declared policy

1. **Why is 0.50 not an automatic deposit threshold?**
   Answer with the decision, the calculation and one failure mode.

2. **What evidence says the predicted probabilities can support a policy?**
   Answer with the decision, the calculation and one failure mode.

3. **Which cost change would move the break-even threshold upward?**
   Answer with the decision, the calculation and one failure mode.

The final threshold must survive calibration, time shift, cost sensitivity and group checks.

Write one hypothesis that can survive a result

Record the hypothesis before running anything.

CHANGE	One parameter, feature decision or modeling choice.
EXPECT	The direction of change, and why.
MEASURE	Validation period, runtime and the required metrics.
DECIDE	Keep, reject or follow up, based on the declared evidence.

EXAMPLE · BOOSTING

Lowering the learning rate from 0.10 to 0.05 with early stopping will reduce later-period log loss, because smaller steps overfit the 2016 demand pattern less.

EXAMPLE · TODAY'S SVM (OPTIONAL)

A calibrated linear SVM will match logistic log loss within 0.005 but not beat it, because both draw linear boundaries; record runtime.

A complete package connects evidence to a deposit decision

COMPONENT	WHAT IT MUST CONTAIN
Notebook	Executed and reproducible, with at least five deliberate experiments
Decision file	booking_id, cancellation_probability, require_deposit
Memo	One page: recommendation, threshold, expected value
Analysis	Calibration, cost sensitivity, time shift, and three operational groups
Disclosure	Document consequential AI-assisted work

Due Sunday, November 15, 11:59 PM ET. Individual exercise.

BLACKBOARD › Exercises › Exercise 03

Interest Exchange complete; proposals now take shape

OCT 27
Register a proposed team
11:59 PM ET

OCT 28
Instructor confirms teams
6:00 PM ET

NOV 2
Class: align the proposal
In class

NOV 8
Proposal + signed contract
11:59 PM ET

WHAT THE PROPOSAL MUST SETTLE

Stakeholder · decision · unit and moment · permitted data · baseline · evaluation · risks and kill criterion

Teams become official only after instructor confirmation. Model choice comes last, not first.

BLACKBOARD › Capstone area

A strong proposal defines the decision before the model

Seven fields, each with the question a reviewer will ask.

FIELD	THE REVIEWER'S QUESTION
Stakeholder	Who acts on the output, and do they want it?
Decision	What changes because of the prediction: which action, how many, how often?
Unit and moment	What receives one prediction, and at what point in time?
Permitted data	Can you get it by Week 10, is it allowed, and is every feature known at the moment?
Baseline	What happens today, and what does a simple rule achieve?
Evaluation	Which metric, on which later data, translated into what decision value?
Risks and kill criterion	Which result ends or reshapes the project?

Rewrite: from a topic to a decision

WEAK

“Use machine learning to predict customer churn for a telecom company.”

No owner, no action, no moment, no baseline, no way to fail.

STRONG

Owner	Retention team lead
Decision	Which customers receive a save call this week, at most 500 calls
Unit · moment	One customer, 30 days before contract renewal
Data	Public IBM Telco Customer Churn data: account, services, charges
Baseline	Call month-to-month customers with tenure under 12 months
Evidence	Net retained value at 500 calls versus the rule; calibration; segment slices
Risk · kill	No event dates, so no true time split: say so. Stop if value gain < 10% over the rule

Two strong proposals: text routing and public-service triage

Each names the action, the moment, data known at that moment, a beatable baseline and a way to fail.

A · SUPPORT-TICKET ROUTING

Decision	Route each new ticket to the right queue within five minutes; a human handles ambiguous cases
Data	Ticket text known at submission (e.g., public Customer Support on Twitter data)
Baseline	Current keyword-routing rule
Evidence	Macro F1 and routing delay on later tickets; language slices; abstention rate
Kill	Gains over the rule disappear on later tickets

B · NYC 311 SLOW-CASE ALERT

Decision	Flag new 311 requests likely to stay open more than 10 days, so supervisors act early; 50 flags a day
Data	NYC Open Data 311 fields known at creation: type, agency, location, time
Baseline	Historical median closure time by complaint type
Evidence	Precision at 50 flags a day on later months; borough slices
Kill	No lift over the complaint-type rule in later months

Two more: an operations alert and a regression problem

Each names the action, the moment, data known at that moment, a beatable baseline and a way to fail.

C · LATE-DELIVERY OUTREACH

Decision	At dispatch, send a proactive delay message to orders likely to arrive late
Data	Public Olist e-commerce orders: seller, customer region, weight, promised date. Exclude actual delivery date
Baseline	Rule on seller–customer distance and promised window
Evidence	Recall at a fixed message volume; calibration; later months
Kill	Precision at 5% flagged no better than the rule

D · NEW-LISTING PRICE GUIDANCE

Decision	Suggest a nightly price range to a new host at listing creation
Data	Inside Airbnb NYC: listing attributes and location. Exclude reviews and occupancy (known only later)
Baseline	Neighborhood × room-type median price
Evidence	MAE on listings created later; error by borough and price band
Kill	MAE gain under 10% versus the baseline

Six proposal red flags to fix before November 8

NO DECISION OWNER

Nobody would act on the output.

TARGET KNOWN ONLY AFTER THE DECISION

The label leaks into features or arrives too late.

DATA YOU CANNOT GET BY WEEK 10

Access, license or privacy blocks the build.

NO BASELINE

There is no way to show the model adds value.

METRIC NOT TIED TO ACTION

Accuracy improves; nobody's decision does.

SCOPE BIGGER THAN FIVE WEEKS

Weeks 10–14 cannot absorb it; cut to one decision.

Near-term work and the course finish

Blackboard controls dates. Complete only released work.

DATE	ITEM	WHERE
Tue Oct 27, 11:59 PM ET	Register proposed capstone team	Capstone area
Wed Oct 28, 6:00 PM ET	Official teams confirmed	Capstone area
Fri Oct 30, 3:00–6:00 PM	Midterm	In class
Sun Nov 8, 11:59 PM ET	Capstone proposal and signed contract	Capstone area
Sun Nov 15, 11:59 PM ET	BookingGuard (Exercise 03)	Exercises
Pending release	Segment Studio and Problem Set 03	Blackboard
After Week 14	Final exam; date to be confirmed	Blackboard

BLACKBOARD › Course outline

Exit ticket: three next actions

1

Register your proposed capstone team

By Tuesday, October 27, 11:59 PM ET

2

Finish the midterm reference sheet

Add today's margin, hinge loss and scaling rules

3

Open BookingGuard and record the first hypothesis

Change · expect · measure · decide

Next class meeting: the midterm, Friday, October 30.

APPENDIX

Optional references

SVM mathematics, kernel functions and further study

Soft-margin SVM: primal and dual

$$\textbf{Primal:}\quad \text{minimize}\ \tfrac{1}{2}\|w\|^2 + C\sum_i \xi_i$$

$$\text{subject to}\quad y_i(w\cdot x_i + b) \geq 1 - \xi_i,\quad \xi_i \geq 0$$

$$\textbf{Dual:}\quad f(x) = \sum_i \alpha_i y_i K(x_i, x) + b,\quad 0 \leq \alpha_i \leq C$$

SLACK ξ_i	How far point i intrudes past its margin edge; $\xi_i > 1$ means misclassified.
$\alpha_i > 0$	Only for support vectors. Every other point drops out of f(x).
$K(X_i, X)$	Data enter only through similarities, so a kernel can replace the dot product.

Common kernels and their parameters

KERNEL	KERNEL FUNCTION	PARAMETERS	USE WHEN
Linear	$x \cdot z$	C	Sparse or wide data; a fast baseline
Polynomial	$(\gamma\, x \cdot z + r)^d$	C, γ, r, degree d	Known low-order interactions
RBF (Gaussian)	$\exp(-\gamma \lVert x - z \rVert^2)$	C, γ	General smooth nonlinear boundary
Sigmoid	$\tanh(\gamma\, x \cdot z + r)$	C, γ, r	Rarely the first choice

scikit-learn default: gamma = “scale” = 1 / (n_features × variance of X). Another reason to standardize first.

Visual explainers and references for SVMs

TEXTBOOK	ISLP, Chapter 9: Support Vector Machines. statlearning.com
USER GUIDE	scikit-learn: Support Vector Machines, including scores and probabilities. scikit-learn.org/stable/modules/svm.html
CALIBRATION	scikit-learn: Probability calibration. scikit-learn.org/stable/modules/calibration.html
INTERACTIVE	SVM classifiers on the iris dataset: compare linear and RBF boundaries. scikit-learn.org/stable/auto_examples/svm/plot_iris_svc.html
INTERACTIVE	RBF SVM parameters: watch C and γ reshape the boundary. scikit-learn.org/stable/auto_examples/svm/plot_rbf_parameters.html
COURSE TOOL	SVM Margin Explorer: add points, move C and γ, toggle scaling. lathrahul.github.io/modern-ai-ml-exercises/week-08/svm-margin-explorer.html
BIAS–VARIANCE	MLU-Explain: The Bias–Variance Tradeoff (interactive). mlu-explain.github.io/bias-variance